

IBEH CHIMCHETAM CHRISTABEL

STC Data Analyst Project - Week 1 Report

Problem:

Massachusetts General Hospital is experiencing a sustained increase in operational costs and significant inconsistency in patient length of stay across its departments. Readmission rates remain high among key patient groups, and it is currently unclear what the primary drivers of these problems are.

Business Questions:

- What is the re-admission rate per 30days?
- Does insurance coverage affect re-admission rate and Length of stay durations within patients with similar diagnosis?
- What demographic group is associated with longer/shorter stay duration and higher re-admission rate?
- What procedure incurs the most cost post insurance coverage?
- What department records the highest case of re-admissions?
- Are there departments that consistently predict longer stays and higher costs across diagnoses?
- What demographic and operational factors are most strongly associated with length of stay above the average for each department?

Stake Holders and What they want to see

Stakeholder 1 — Hospital CEO

Role: Oversees the entire institution's performance and strategic direction.

Expected Insights:

Cost trends, average length of stay, readmission rate over time

Stakeholder 2 — Chief Financial Officer (CFO)

Role: Manages hospital budget, billing, insurance contracts, and cost control.

Expected Insights:

Cost per patient by insurance type, diagnosis, and department

Quantified savings opportunities tied to specific interventions

Insurance coverage vs. actual cost gaps

Stakeholder 3 — Chief Medical Officer (CMO)

Role: Responsible for care quality, clinical protocols, and patient outcomes.

Expected Insights:

Readmission rates by diagnosis and department

Drivers of extended stay periods

Risk-adjusted analysis that protects patient safety

Stakeholder 4 — Operations Manager

Role: Manages day-to-day workflow; bed allocation, staffing, scheduling, and discharge planning.

Expected Insights:

Which wards or departments have the longest average stays

Bed occupancy rates and bottlenecks by ward and time period

Patterns linking discharge delays to specific days or shift patterns

Data enabling predictive bed planning